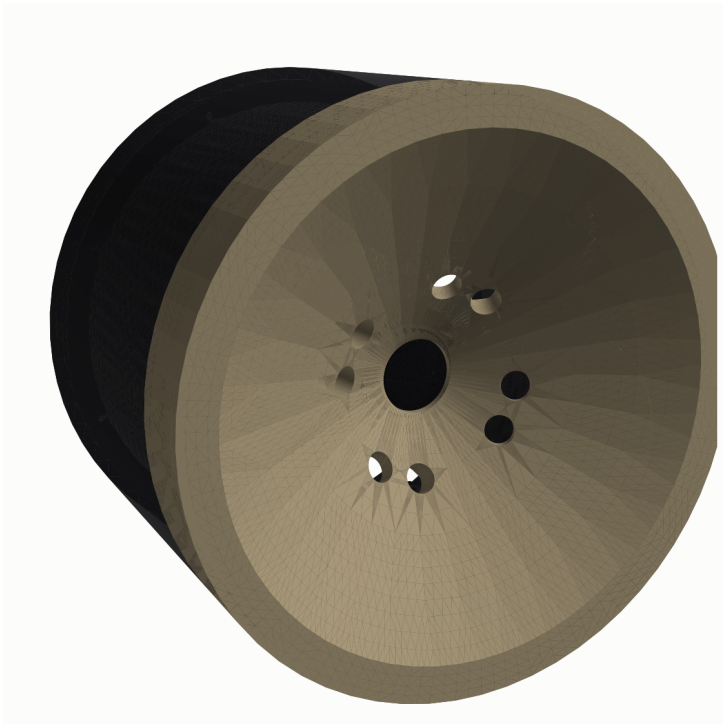


Circular MEH Bluetooth Speaker

Design package - a 170 mm, exactly circular multiple-entry horn (Synergy-style point source): textile-dome tweeter at the horn apex, four tiny mids injecting through ports in the horn wall, sealed back, passive-radiator bass, DSP-active electronics.



Front horn module. The eight small openings around the throat are the mid-driver injection ports (two per driver); the tweeter sits at the apex behind the throat plate.

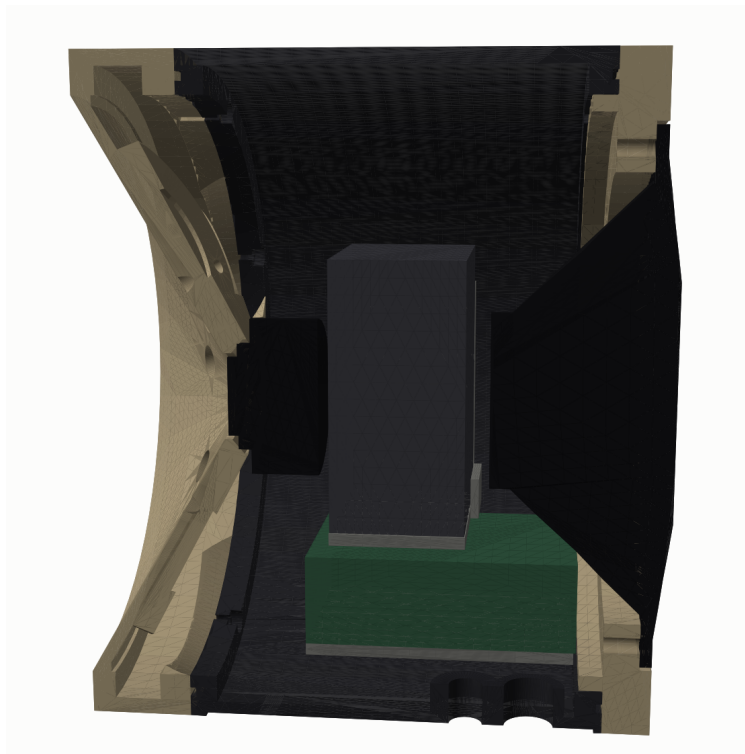
Headline numbers

Attribute	Value
Envelope	Ø170 mm face x 148 mm long (perfect circle, cylindrical barrel)
Topology	2-way MEH: 4 mids into 120° conical horn + dome tweeter at apex; shared sealed rear chamber vented by rear-firing passive radiator
Drivers	4x Dayton DMA45-4 (1.5") - Dayton ND20FB-4 (3/4" textile dome) - Dayton DSA135-PR (5" passive radiator)
Electronics	Dayton KABD-230 (2x30 W, ADAU1701 DSP, BT 5.0 aptX HD) + LBB-5Sv2 battery board, 5x18650 (21 V)
Crossover	3.4 kHz LR4 (DSP) reinforced by the injection-port acoustic low-pass ($f_0 \approx 4278$ Hz)
Bass tuning	$F_b = 76$ Hz (net $V_b \approx 1.60$ L, +4 g tuning mass on the PR)
Max output	≈ 98 dB @ 1 m midband; 92-98 dB through the 80-200 Hz range; usable bass to ~ 75 Hz with DSP shelf
Build	4 printed parts (STEP + STLs supplied), 8 heat-set M3 joints, 2 O-ring gaskets, $\sim$ \$250 in parts

1. How this MEH works

A multiple-entry horn makes all drivers behave as one point source: the tweeter radiates from the horn apex, and the four mids inject through small ports in the horn wall close enough to the apex that, at the crossover frequency, mids and tweeter are acoustically coincident. The classic rule places the injection ports within a quarter wavelength of the tweeter at crossover. Here the acoustic path from the tweeter exit to the port centres is 24.5 mm against a quarter-wave budget of 25.2 mm at 3.4 kHz - inside the rule, so the crossover sums cleanly with no lobing.

Each mid is bolted to a pod on the back of the horn cone and fires through a shallow front chamber (a 1.2 mm dish under the cone) into two 9 mm ports. Chamber plus ports form a 2nd-order acoustic low-pass at about 4.3 kHz - above the crossover, where it adds free out-of-band rejection and keeps horn-interior disturbance small. The mids' rear waves share one sealed chamber, which drives the 5" passive radiator on the back panel: the back of the speaker stays completely sealed (a PR is airtight), and bass below the mids' reach is radiated by the PR around its 76 Hz tuning.



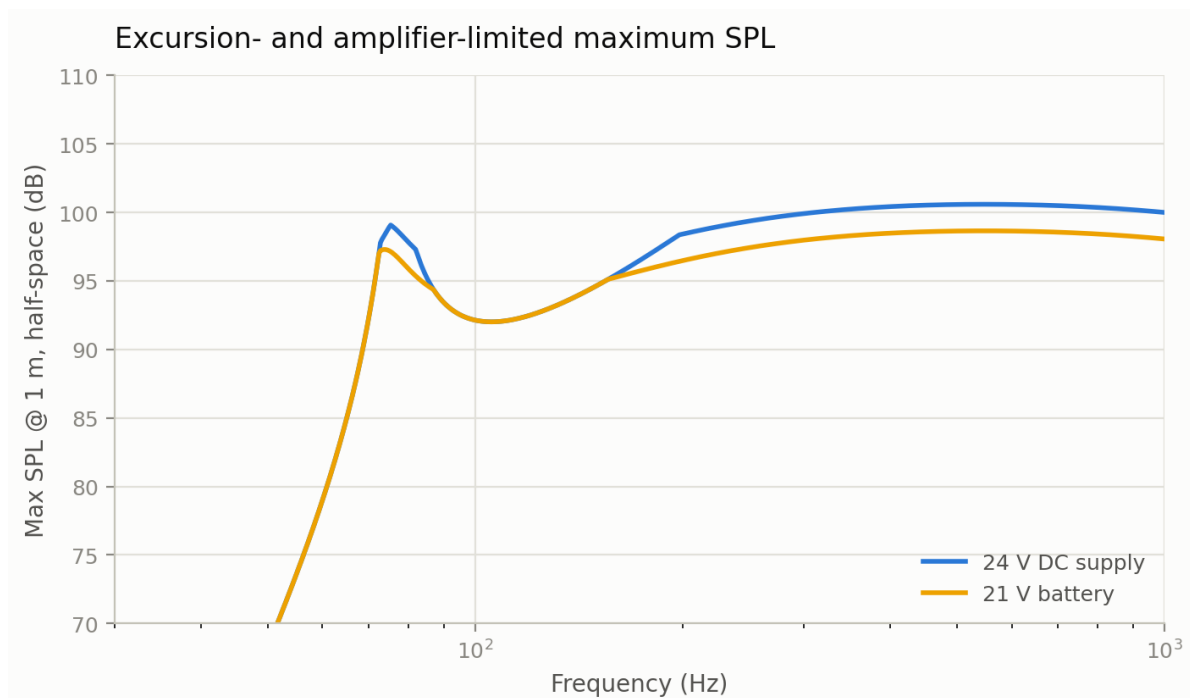
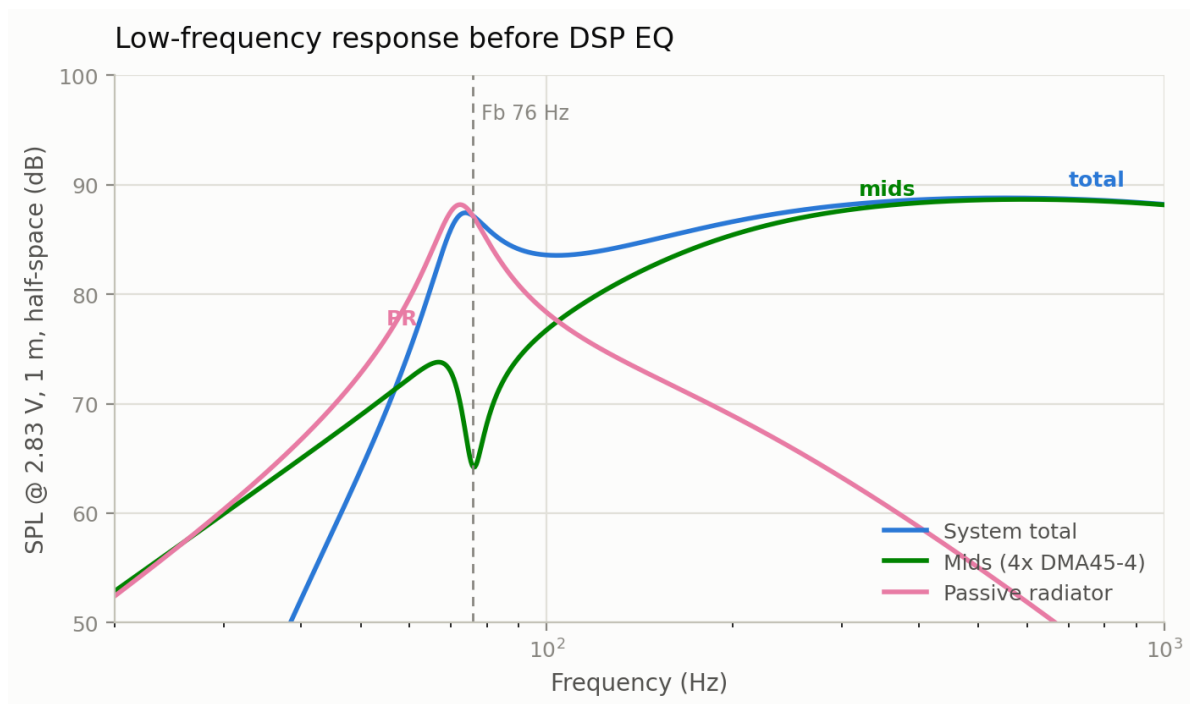
Cross-section. Left: horn face with a mid pod and injection ports; centre: standing battery pack behind the tweeter, KABD-230 amp (green) on the bottom tray; right: passive radiator in the rear ring.

Horn and injection geometry

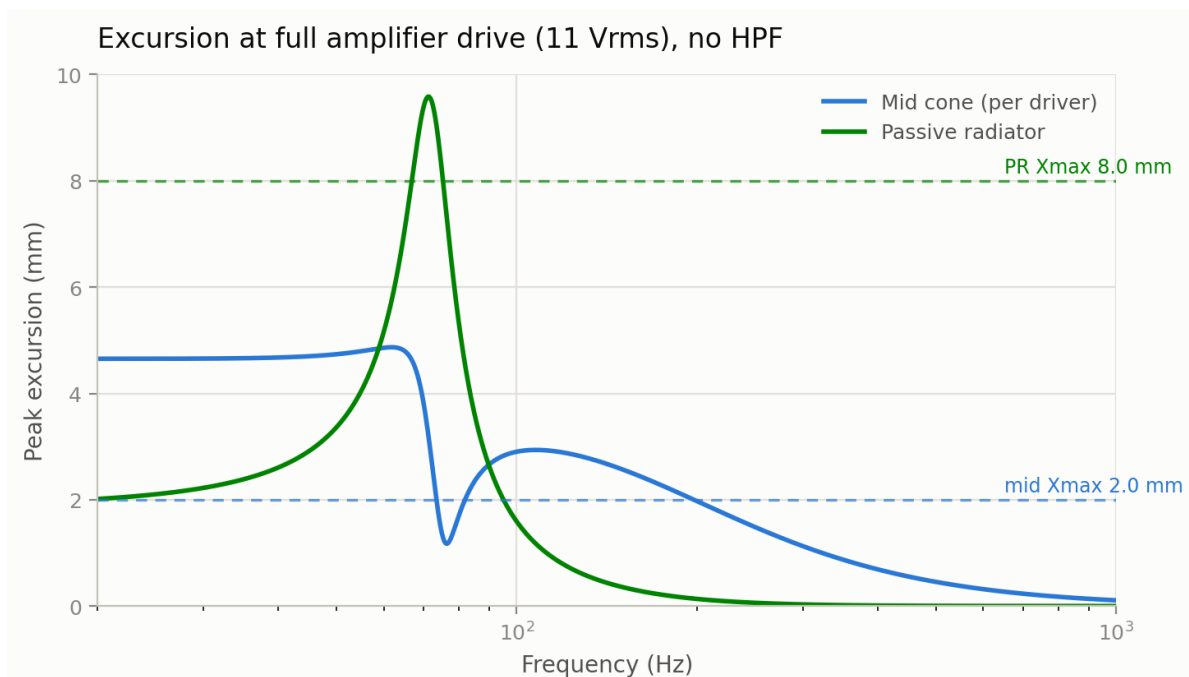
Parameter	Value	Parameter	Value
Coverage angle	120° conical	Mouth / throat dia	148 / 22 mm
Horn depth	36.4 mm	Tweeter	rear-mounted behind Ø29 throat plate
Mid ring radius	39.5 mm (slant 45.6 mm)	Driver tilt	axes 30° off the speaker axis
Ports per mid	2 x Ø9 mm	Port length	5.3 mm (wall thickness)
Port path from throat	24.5 mm	$\frac{1}{4}\lambda$ at crossover	25.2 mm
Front chamber	Ø41 x 1.2 mm dish (≈1.6 cc)	Port bandpass f0	≈4.3 kHz

2. Bass alignment and output limits

The four mids double as the bass engine. Their combined rear wave pressurises the 1.60 L sealed chamber and drives the DSA135-PR. With the passive radiator's stock moving mass plus 4 g of added mass (washers on its M5 stud), the box tunes to 76 Hz - independently confirmed in the lumped-element simulation by the mid-cone excursion minimum at 76-77 Hz. Below are the modelled responses before any EQ; the DSP then applies a low shelf to flatten the 80-200 Hz region at the expense of maximum level.



Maximum SPL is limited by mid-driver excursion (2 mm X_{max}) through the midbass and by the 30 W amplifier above ~200 Hz. The passive radiator (8 mm X_{max}) never limits first. On battery (21 V) the amp ceiling drops ~2 dB; everything else is unchanged.

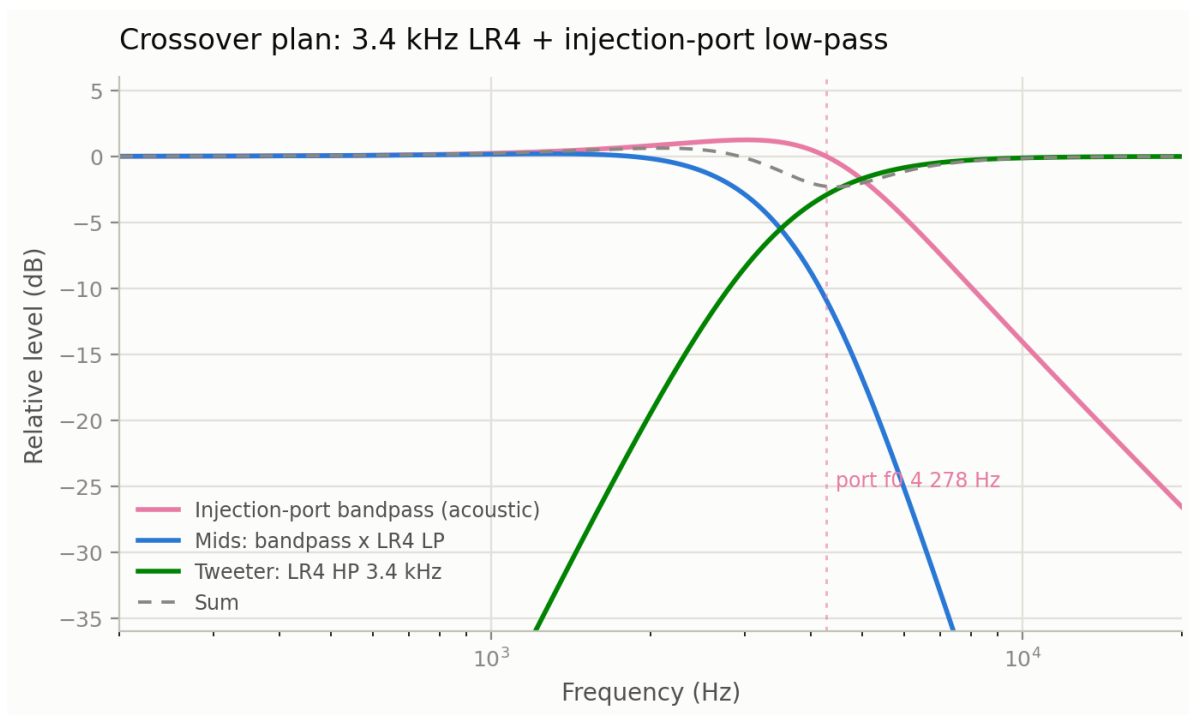


Excursion at full drive with no high-pass: the mids reach Xmax near 100-130 Hz, which is what the DSP limiter and 55 Hz protective high-pass are set around. The excursion null at 76 Hz is the classic passive-radiator signature and confirms the tuning.

Retuning by ear or measurement

The PR's threaded stud accepts washers: no mass gives $F_b \approx 82$ Hz (tighter, punchier), 4 g gives the designed 76 Hz, and ~8-10 g reaches ~70 Hz (deeper, less midbass headroom). Retune, re-measure the impedance minimum, and adjust the DSP shelf to match.

3. Crossover and DSP programme



Starting settings for SigmaStudio on the KABD-230 (final values should be tweaked against a measurement mic):

Channel	Block	Setting
Ch 1 - mids (2S2P, 4 Ω)	High-pass	55 Hz Butterworth 12 dB/oct (excursion protect)
	Low shelf	+6.5 dB below 150 Hz (flattens response to ~78 Hz)
	Low-pass	LR4 at 3.4 kHz
	Limiter	threshold $\approx$ -4 dBFS, attack 5 ms, release 200 ms
Ch 2 - tweeter (4 Ω)	High-pass	LR4 at 3.4 kHz
	Level	-3.5 dB pad (90 dB tweeter vs ~86.6 dB mid array)
	Delay	start at 0.00 ms; scan $\pm$ 0.05 ms for best null when one channel is inverted, then revert polarity
Both	Bass enhance	optional KABD preset OFF - the shelf above replaces it

4. Bill of materials

Part	Role	Qty	Price (MSRP)
Dayton Audio DMA45-4 1-1/2" full-range	mid drivers (bass + mid duty)	4	\$11.99 ea (\$8.98 street)
Dayton Audio ND20FB-4 3/4" textile dome, rear-mount	HF at horn apex	1	\$16.99
Dayton Audio DSA135-PR 5" aluminium passive radiator	bass radiator	1	\$27.99
Dayton Audio KABD-230 2x30 W DSP + BT 5.0 amp board	amp / DSP / Bluetooth (KABD-250 is a drop-in upgrade if 230 is out of stock)	1	~\$70
Dayton Audio LBB-5Sv2 5S charger/battery board	21 V battery + charging	1	\$41.99
18650 Li-ion cells (≥ 2600 mAh, e.g. Samsung 26J)	battery	5	~\$25
M3 heat-set inserts + M3/M4 screws, O-ring cord $\varnothing 2$ mm, foam gasket tape	assembly + sealing	-	~\$15
DC barrel jack (5.5x2.1), latching power button	IO panel	1+1	~\$5
PETG/PLA filament	4 printed parts, ~700 g total	-	~\$15
Total			≈ \$250-265

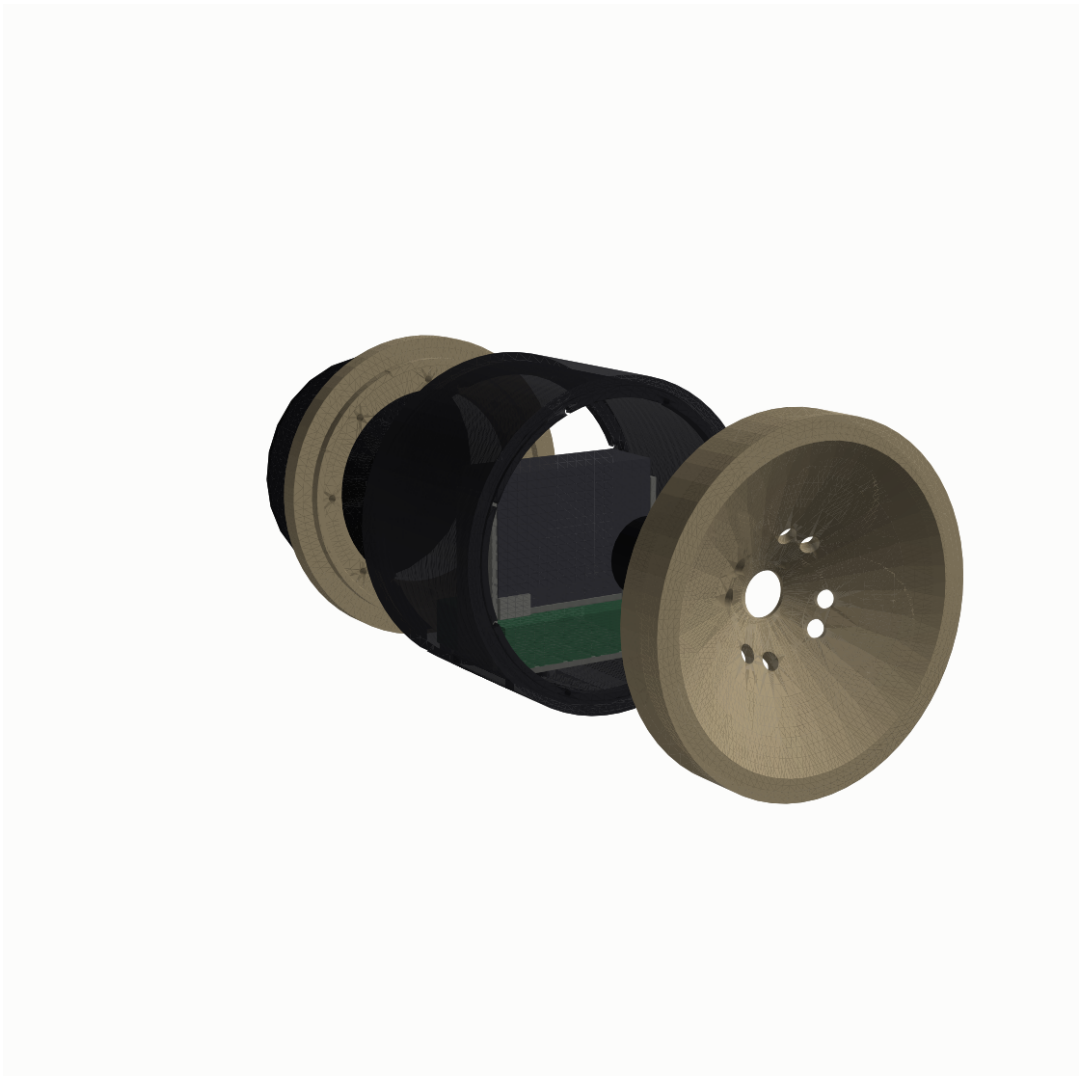
Wiring

Mids in two series pairs, pairs in parallel (2S2P) to Ch 1: $4\ \Omega$ nominal, 40 W handling against the 30 W channel. Tweeter direct to Ch 2. Battery board output to the KABD J6/J8 power input; the DC jack feeds the LBB-5Sv2 charge input so the speaker plays while charging. Pick opposite mids for each series pair (top-left with bottom-right) so any pod-to-pod variation averages out.

Verify against your parts before printing

Modelled from published data sheets; three dimensions are worth checking with calipers because vendors have been known to revise tooling: the DMA45-4 corner-hole bolt circle (modelled $\varnothing 50$ mm), the DSA135-PR mounting-hole circle (modelled $\varnothing 123$ mm, 8 holes), and the KABD-230 standoff pattern (the tray is drilled on assembly - transfer-mark through the board). All are one-line edits at the top of meh_cad.py; re-running it regenerates every file.

5. Printing and assembly



Exploded: horn module (with mids on their pods and the tweeter behind the throat), barrel shell with tray, battery and amp, rear ring, passive radiator.

Step	Detail
Print	horn module face-down with tree supports under the pods; shell vertical, no supports; rear ring and tray flat. PETG, 0.2 mm layers, 5 perimeters, $\geq 25\%$ gyroid - perimeters matter more than infill for airtightness. Brush thinned epoxy or 3D-Gloop on interior seams.
Inserts	M3 heat-set: 8 in the horn joint ring, 8 in the rear ring, 16 in the mid pods (4 per pod), 4 in the tweeter bosses, 4 in the tray bosses.
Drivers	Tweeter first (from inside the open shell-less module), foam gasket under its flange. Then the four mids onto their pods (gasket tape under each frame), dressing each pair's wiring toward the bottom.
Electronics	Battery into the cradle (cable-tie over the top), amp on the tray, tray onto its bosses. DC jack and button into the bottom holes; wire before closing.
Close	O-ring cord into the front groove - shell onto the horn module - 8 screws from inside the rear opening. O-ring into the rear groove - rear ring on - 8 screws. PR last, onto the rear ring with its gasket, 8 M4 screws. Every fastener is reachable through the PR cutout.

Step	Detail
Tune	Measure impedance: the twin peaks should straddle ~76 Hz. Add or remove PR washers to move Fb, then load the DSP programme.

Verification summary (automated checks on this exact geometry)

All four printed parts are watertight, manifold meshes. Programmatic clearance checks between every driver, board and structure pair found no interference (worst true clearance in the packed bottom bay: amp-to-PR basket 2.0 mm; neighbouring mid frames 1.4 mm; battery to PR hub 6.1 mm). PR tuning cross-checks: formula 76.0 Hz vs simulation 76.5 Hz. Injection-port path $24.5 \text{ mm} \leq$ quarter-wave budget 25.2 mm; port bandpass 3.8 kHz sits above the 3.4 kHz crossover. CAD-measured net chamber volume 1.65 L vs 1.60 L used in the simulation (wiring allowance).

Files in this package

meh_speaker_assembly.step - full CAD assembly with driver/board placeholders (import into Fusion, SolidWorks, FreeCAD). horn_module.stl, shell.stl, rear_ring.stl, tray.stl - print-ready. meh_cad.py - the parametric source (CadQuery); every dimension in this document is a named constant at the top. acoustics.py - the simulation that produced the charts; edit and re-run if you change drivers or volume.

Data sources: Dayton Audio / Parts Express published specifications (DMA45-4, ND20FB-4, DSA135-PR, KABD-230/250 manual, LBB-5Sv2), loudspeakerdatabase.com T/S entries. Lumped-element PR model after Small (1974); MEH port placement per the Danley Synergy quarter-wave practice.